

Console Tester technical reference

This is the authoritative technical description of the released and hardware-validated Public Beta 0.8 diagnostic.

Address map

CPU address	Hardware or content	Notes
>0000->1FFF	Console CPU ROM	Read twice for stability and identity
>6000->7FFF	Diagnostic extension	Cartridge header, menu, and interactive tests
>8300->83FF	Common scratchpad	Saved, tested, and restored
>8400	Sound write port	Muted at startup; driven by the PSG test
>8800 , >8802	VDP read/status	Direct hardware access
>8C00 , >8C02	VDP write/control	Direct hardware access
>9800 , >9C02	GROM read/address	Stable-read and identity checks only
>C000->DFFF	Tester U7 SRAM	Workspace, results, and relocated demo code
>E000->FFFF	Fixed diagnostic core	Automatic tests and LOAD recovery
>E002 on writes	Six-bit LED latch	Direction decode separates writes from ROM reads
>FFFC->FFFF	LOAD vector	Workspace >C000 , entry >FF00

The ordinary TI 32K expansion also drives >C000->FFFF and must not be present while the tester owns U7 SRAM and U3 ROM. The raster controller runs from the tester's own U7 SRAM at >D000->D537 ; it does not require optional RAM at >A000 .

W27C512 layout

The tested 16 KiB modification is direct mapping, not bank switching. The 65,536-byte programmer file is sparse:

File offsets / CPU addresses	Content
>6000->7FFF	8 KiB interactive extension
>E000->FFFF	8 KiB automatic core
all other offsets	>FF

The fixed core validates the extension signature, version, entry range, ABI, and zero word sum before calling it. A missing or corrupt extension cannot prevent the automatic tests, LED report, LOAD path, or fixed-core result page.

The extension begins with a standard TI-99/4A cartridge header. Its menu entry, HEXBUS DIAG 1.0 , executes BLWP @>FFFC , the same vector used by the physical LOAD switch. A TI-99/4 does not use that cartridge menu and starts through LOAD.

Automatic test coverage

Target	Coverage
TMS9900	Representative arithmetic, logic, shift, compare, and branch paths
Tester SRAM	>C100->DFFF , excluding active workspace/result storage
Scratchpad	>8300->83FF , save/test/restore
VRAM	All >0000->3FFF bytes with 00 , FF , AA , and 55
Console CPU ROM	>0000->1FFF , forward and reverse reads
Diagnostic core	Complete >E000->FFFF zero word sum
System GROMs	GROM 0, 1, and 2, two 6 KiB reads each
LED latch	Six-output walk followed by settled PASS/FAIL groups

The VRAM test is destructive. A failure or power interruption cannot promise that scratchpad state will be restored.

Firmware identity and read health

Read health and firmware identity are deliberately separate. Two reads that do not agree are a diagnostic failure. A stable custom, replacement, or mixed ROM set is reported as information and is not colored or labeled as failed.

Profile	CPU ROM word sum	GROM 0 byte sum	GROM 1	GROM 2
TI-99/4 (1979)	>F1BB	>B16C	>1B7B	>9EC3
TI-99/4A (1981)	>7D8C	>5574	>1B99	>9ADC
TI-99/4QI V2.2 (1983)	>7D8C	>31D7	>1B99	>9ADC

The CPU value is the modulo-65536 sum of 4096 big-endian words. Each GROM value is the modulo-65536 sum of 6144 bytes. These compact runtime identifiers are not catalog-file CRC or SHA hashes.

Keyboard and joystick ownership

The diagnostic does not call the console ROM keyboard vector at >000E and does not call a GROM routine. It scans the TMS9901 directly, using separate translation tables for the five-column TI-99/4 keyboard and six-column TI-99/4A keyboard. Unknown or GROMless recovery systems default to the /4A table so menu input remains available.

The TI-99/4 uses joystick selector columns 5 and 6. The TI-99/4A and QI use columns 6 and 7. The page reports modifier state separately from translated characters, recognizes two-key chords plus Alpha Lock, and never executes QUIT or other console actions while the diagnostic is active.

Alpha Lock is advisory rather than a joystick lockout. Both joystick selectors are always scanned and displayed. On the /4A-family path, Alpha Lock down adds JOYSTICKS: RELEASE ALPHA LOCK because the original /4A wiring can interfere with joystick Up. A QI may continue to show valid Up input. The /4 path omits the message because it has no Alpha Lock. The switch probe and joystick reads remain independent so joystick Up cannot by itself create the advisory.

P5 state during the joystick reads is intentional. With Alpha Lock released, the /4A-family path holds P5 low across both selector reads. Testing on a GROMless /4A showed that restoring P5 high masked the shared INT7 Up input on both joystick ports. With Alpha Lock down, P5 remains high so the switch cannot be mistaken for Up. The TI-99/4 path also retains its normal P5-high state.

Video and sprite contract

The TI-99/4 skips the Graphics-II bitmap stage because its original TMS9918 does not support that mode. Text, character, Multicolor, sprite, and raster tests remain available. Operator-facing screens say `VDP`; they do not infer a specific VDP model from display behavior.

The sprite exercise writes five same-scanline entries to the sprite attribute table at `>0300`, using pattern table `>1800`. Red sprite 0 and white sprite 1 exactly overlap for the collision condition. Red has priority, so the pair looks like one clean red object instead of a damaged composite. Cyan and light-blue sprites 2 and 3 are separate. Yellow sprite 4 exercises the four-per-line limit.

At the original four-sprites-per-scanline limit, the expected display is three visible groups: the red collision pair, cyan, and light blue. Yellow is suppressed and the fifth-sprite flag is set. An F18A or Pico9918 configured for an expanded limit may display yellow. Collision and fifth-sprite flags remain observations, not independent pass/fail claims.

Every return from a video exercise blanks the display and resets the VDP control latch. It then erases both sprite attribute areas owned by the tester: the diagnostic SAT at `>0300` and the MegaDemo timing SAT at `>3800`. After restoring all eight UI registers, it rebuilds the font, colors, and frame, terminates both SATs again, and only then enables display. This ownership boundary prevents demo sprites or table selections from corrupting later pages.

Enhanced VDP identification

The startup screen identifies an enhanced programming interface without claiming that software can distinguish every original TMS die. A legacy result is deliberately shown as `TMS FAMILY`, not specifically `TMS9918` or `TMS9918A`.

The probe follows Troy Schrapel's published F18A/Pico9918 method:

1. Disable display interrupts in VDP register 1, then write `>1C` twice to enhanced register 57 to request unlock.
2. Restore register 1 and place the six-byte GPU program `04 E0 3F 00 03 40` at VRAM `>3F00`. This is `CLR @>3F00; IDLE`.
3. Start the GPU at `>3F00`, wait for at most one bounded frame interval, and read that VRAM byte. A legacy TMS VDP only stores the bytes, leaving `>04`; a compatible GPU executes them and changes the byte to zero.
4. Only after that positive execution proof, read enhanced status register 1 for the implementation signature. Restore the status selector to register 0.
5. Reset and relock the enhanced interface through register 50, then restore all eight legacy registers before the destructive VRAM test and normal UI.

Status 1 masked with `>E8` equal to `>E8` identifies Pico9918. Otherwise, status 1 masked with `>E0` equal to `>E0` identifies a genuine F18A. Another GPU-compatible response is displayed conservatively as `F18A COMPAT`. The method and original CVBasic example are documented in [Troy Schrapel's AtariAge post](#) and the [Pico9918 project](#).

The diagnostic deliberately does not display status register 14 as a device firmware version. On both F18A and Pico9918 it reports the F18A-compatibility version. Reading a Pico9918's complete firmware and hardware versions requires the Pico9918 configuration-register interface and is outside this identity test.

The two register-57 writes are intentionally emitted as complete direct VDP control-port transactions. The shared register helper byte-swaps its input as part of normal use, so reusing one loaded value for two helper calls would not send two identical writes and would silently prevent the unlock.

Beta 0.8 qualification

The exact public programmer image passed the complete static acceptance suite, including checks of all 16 AORG/XORG regions, both ROM checksums, the full font, keyboard tables, VDP probe transactions, sprite table, and W27C512 composition. It was then exercised on stock TI-99/4A, TI-99/4QI V2.2, GROMless TI-99/4A LOAD recovery, and a TI-99/4 fitted with a Pico9918. VDP observations included a TMS9918-family device, an F18A, and a Pico9918. This remains a public beta because the sample of original and replacement hardware is necessarily limited.